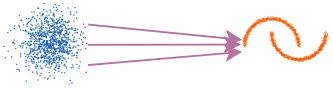
Flow Matching

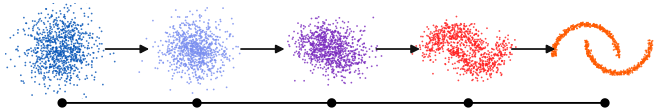
Learn velocity field that transports p_0 to p_1



$$v^*(u_\tau, \tau) = \tau(u_1 - u_0) + u_0$$

Stochastic Interpolant Process

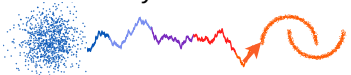
$$u_\tau = I(\tau, u_0, u_1) + \gamma(\tau)\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$



Marginal distribution $p_\tau, \tau \in [0, 1]$

Score-Based Generative Models

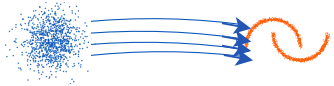
Use score $\nabla_u \log\ p_\tau(u_\tau)$ to define dynamics



$$du_\tau = [f_\tau(u_\tau) - \frac{1}{2} g_\tau^2 \nabla_u \log\ p_\tau(u_\tau)]d\tau + g_\tau dW_\tau$$

Continuous Normalizing Flows

deterministic, invertible probability flow w/o noise



$$du_\tau/\tau = v_\tau(u_\tau, \tau)$$

Diffusion Models

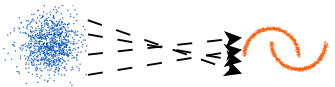
SDE with fixed forward noise schedule



$$du_\tau = f_\tau(u_\tau)d\tau + g_\tau dW_\tau$$

Entropy-Regularized Optimal Transport

Transport mass between p_0 and p_1 with controlled stochasticity $\epsilon > 0$



Schrödinger Bridges

Most likely stochastic process between p_0 and p_1



$$du_\tau = [f_\tau(u_\tau) + g_\tau^2 \nabla_u \log \varphi_\tau(u_\tau)]d\tau + g_\tau dW_\tau$$